

Term paper on Targeted Research Synthesis

Recent Advances and Remaining Gaps in Artificial Intelligence for Cancer Drug Repurposing

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Abstract

Artificial intelligence (AI) is becoming an important tool in cancer drug repurposing. Drug repurposing means finding new cancer uses for already approved or previously studied drugs. This approach can reduce time, cost, and risk compared with traditional drug discovery. However, many AI-predicted drug candidates are still not clinically reliable because they are often based on limited datasets, incomplete drug-target information, and weak biological validation. This paper introduces the topic, compares twenty related studies, and concludes that future AI-based drug repurposing should focus on explainable, patient-specific, and experimentally validated pipelines.

Keywords: Artificial intelligence, cancer, drug repurposing, machine learning, deep learning, precision oncology, drug sensitivity.

Introduction

Cancer drug discovery is slow, expensive, and risky. Developing a completely new drug can take many years, and many drug candidates fail before reaching patients. Because of this, drug repurposing has become an important research direction. Drug repurposing means finding new cancer uses for drugs that already exist. These drugs may already have known safety, dosage, and toxicity information, so they may move faster into clinical testing than brand-new drugs. Recent studies show that artificial intelligence can help this process by searching large datasets, finding hidden drug-disease links, predicting drug sensitivity, and suggesting possible new uses for approved drugs [1]-[5].

Artificial intelligence is useful in this area because cancer research produces many types of data. These include gene expression data, drug-target interaction data, protein interaction networks, clinical trial information, molecular structures, and single-cell data. A human researcher may find it difficult to compare all these data manually. AI and machine learning models can process these data together and identify patterns that may point to useful repurposed drugs [5], [6].

However, cancer is not one simple disease. Tumors differ between cancer types, between patients, and even between cells inside the same tumor. This makes AI-based drug repurposing difficult. A model may predict that a drug will work, but the drug may fail because the model was trained on limited cell-line data, incomplete drug-target information, or bulk tumor data that hides rare resistant cell groups [9], [12].

Recent work is trying to solve this problem by moving from simple prediction toward more patient-specific and biologically meaningful models. For example, DrugReSC uses single-cell transcriptomic data to focus on disease-critical cell subpopulations [12]. CHANCE predicts patient-specific anticancer potential of non-oncology drugs using genome-derived pathways [15]. ResisenseNet uses a hybrid neural network to predict breast cancer drug sensitivity and identify possible repurposed drugs [20]. These newer approaches are promising, but the field still has one major problem: AI can predict many candidate drugs, yet stronger validation is needed before these predictions can be trusted in real clinical practice.

Comparison Table

The following table summarizes twenty recent studies on AI-based cancer drug repurposing published between 2021 and 2026. The papers are compared based on their primary methods, key contributions, and remaining research gaps. This comparison highlights current research trends, identifies common limitations, and provides the foundation for discussing future directions in AI-assisted cancer drug repurposing.

Table I
Comparison of 20 Papers Related to AI-Based Cancer Drug Repurposing

No.	Paper	Year	Main Method	Main Contribution	Main Gap
1	Capistrano et al. [1]	2022	Literature and clinical trial search	Found candidate drugs for cervical cancer repurposing.	Not AI-based; useful mainly as background.
2	Tanoli et al. [2]	2021	ML review and drug-target profiles	Explains AI/ML use in cancer drug repurposing.	Older than 2023-2026; use mainly as background.
3	Malla et al. [3]	2024	Review of computational and experimental methods	Shows repurposing value for resistant and metastatic cancers.	Broad review, not only AI-focused.
4	Xia et al. [4]	2024	Review of mechanisms, organoids, and combinations	Connects repurposing with tumor microenvironment and validation.	Very broad; not only computational.
5	Herraiz-Gil et al. [5]	2025	ML, DL, and AI workflow review	Summarizes AI tools for cancer drug repurposing.	Notes data quality and interpretability issues.
6	Garg et al. [6]	2024	AI-CADD, docking, QSAR, ADMET	Shows how AI improves anticancer drug discovery.	More about drug development than repurposing.
7	Cao et al. [7]	2024 /2025	Structure-based, network-based, and ML methods	Gives a clear map of computational repositioning methods.	Review only; no new model tested.
8	Raju et al. [8]	2025	Broad AI oncology review	Links AI with diagnosis, therapy, and drug development.	Repurposing is only one part.

9	Dara and Ahmed [9]	2026	AI and multi-omics	Connects genomics, proteomics, metabolomics, and repurposing.	Data heterogeneity and bias remain issues.
10	Yu et al. [10]	2023	I-SPY2 and CMap signature reversal	Found fulvestrant as a possible resistance-reversing drug.	Limited laboratory validation.
11	Ahmed et al. [11]	2023	Network medicine and DTI	Found possible drugs and combinations for HPV cervical cancer.	Mostly computational validation.
12	Liu et al. [12]	2024	Single-cell RNA-seq and random forest	Targets disease-critical cell subpopulations.	Needs more clinical validation.
13	Sengupta et al. [13]	2024	QSAR and SVM	Built colorectal cancer models and suggested FDA-approved candidates.	In silico predictions need laboratory testing.
14	Neelima et al. [14]	2026	AutoQSAR, docking, MD, and experiments	Found possible tubulin binders such as omeprazole and podofilox.	Focused only on tubulin.
15	Dong et al. [15]	2025	Supervised ML and genome-derived pathways	Predicts patient-specific anticancer potential of non-oncology drugs.	Needs larger clinical validation.
16	Lee et al. [16]	2024	Ex vivo screening and interpretable ML	Finds repurposable neuroactive drugs for glioblastoma.	Cancer-type specific.
17	Huang et al. [17]	2024	Graph foundation model	Performs broad zero-shot drug repurposing.	Not cancer-specific, but method is relevant.
18	Picard et al. [18]	2024	Multi-layer networks and ML	Predicts new therapeutic targets for prostate cancer.	Predicts targets more than direct drug candidates.
19	Lenhof et al. [19]	2024	Conformal prediction and drug prioritization	Adds confidence/reliability to anticancer drug sensitivity prediction.	Supports repurposing indirectly.

20	Karampuri et al. [20]	2024	1D-CNN, LSTM, and DNN	Predicts breast cancer drug sensitivity and identifies possible repurposed drugs.	Needs future molecular dynamics and in vitro/in vivo validation.
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Conclusion

AI has made cancer drug repurposing faster and more systematic. Instead of depending only on chance discoveries, researchers can now use machine learning, deep learning, network medicine, transcriptomic signature reversal, molecular docking, single-cell analysis, and graph-based models to search for new uses of existing drugs. The comparison table shows that these methods are being used across several cancer types, including cervical cancer, breast cancer, colorectal cancer, glioblastoma, melanoma, renal cell carcinoma, and prostate cancer.

The most important lesson from the reviewed papers is that prediction alone is not enough. Many studies can produce a list of possible drug candidates, but fewer studies prove that those candidates work in patient-like systems. Some studies depend heavily on public databases, cell lines, or computational scores. Others are stronger because they include experimental validation, but they may focus on only one cancer type, one target, or one dataset.

The main research gap is therefore the prediction-to-clinical-validation gap. In simple words, AI is good at suggesting drugs, but the field still needs stronger proof that these drugs will work safely and effectively in real patients. Future research should combine multiple methods instead of using only one model. A strong pipeline should include network medicine, gene-expression reversal, single-cell analysis, drug-target prediction, toxicity prediction, uncertainty estimation, and validation using patient-derived organoids or resistant cancer cell models.

Overall, AI-based cancer drug repurposing is a promising direction for precision oncology. It may help researchers find cheaper, faster, and more personalized treatment options, especially for patients with resistant cancers. To reach this goal, future models must be explainable, tested on diverse patient data, and validated experimentally before moving toward clinical trials.

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